SEPARATED GROUP SPACES OVER A FIELD ARE SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. The author flags the sibling proof (spaces-over-fields lemma used for the lft case) as recommended reading; the proof itself is complete in-span with no omission markers. Cross-chapter same-name note: no same-label card exists in any harvested chapter; the proposition is the general form of the locally-finite-type case carded at L26664's sibling batch position (lemma-group-space-scheme-locally-finite-type-over-k was rejected as a pure three-way reference list, so no double count arises).

Allowed standard prerequisites (editorial).

- Group space over the algebraic closure. lemma-group-space-scheme-over-kbar (same chapter, carded L26646). G_{k-bar} is a scheme, the anchor for the descent chain.
- When a space becomes a scheme after base change. Spaces over Fields lemmas scheme-after-purely-transcendental-base-change, scheme-after-purely-inseparable-base-change, when-scheme-after-base-change, base-change-by-Galois, when-quotient-scheme-at-point (cross-chapter, named block). The four-step field-extension chain and the Galois descent that closes the proof.
- Limits of affine opens. Limits, lemmas descend-opens and limit-affine (cross-chapter, named). Recover W as the base change of an affine open U' over a finite subextension.

Source reference. Original reference 09DT: focused-seeds/geometry/S9-smg-group-space-scheme-over-field/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0B7U: focused-seeds/geometry/S9-smg-group-space-scheme-over-field/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0B8F: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 1296–1296. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0B8E: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 1258–1258. Navigation pointer only: the counted

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Source reference. Original reference 01Z4: focused-seeds/geometry/S9-smg-group-space-scheme-over-field/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01Z6: focused-seeds/geometry/S9-smg-group-space-scheme-over-field/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0B86: focused-seeds/geometry/S9-smg-group-space-scheme-over-field/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0B83: focused-seeds/geometry/S9-smg-group-space-scheme-over-field/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0B85: focused-seeds/geometry/S9-smg-group-space-scheme-over-field/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0B88: focused-seeds/geometry/S9-smg-group-space-scheme-over-field/ref-tags.txt, source lines 12–12. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0B87: focused-seeds/geometry/S9-smg-group-space-scheme-over-field/ref-tags.txt, source lines 13–13. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0B84: focused-seeds/geometry/S9-smg-group-space-scheme-over-field/ref-tags.txt, source lines 14–14. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0479: focused-seeds/geometry/S9-smg-group-space-scheme-over-field/ref-tags.txt, source lines 15–15. Navigation pointer only:

the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Proposition 0.1. *Let k be a field. Let G be a group algebraic space over k. If G is separated, then G is a scheme.*

Proof. This lemma generalizes Lemma 0B8F (which covers all cases one cares about in practice). The proof is very similar to the proof of Spaces over Fields, Lemma 0B88 used in the proof of Lemma 0B8F and we encourage the reader to read that proof first.

By Lemma 0B8E the base change $G_{\overline{k}}$ is a scheme. Let K/k be a purely transcendental extension of very large transcendence degree. By Spaces over Fields, Lemma 0B85 it suffices to show that G_K is a scheme. Let K^{perf} be the perfect closure of K. By Spaces over Fields, Lemma 0B83 it suffices to show that $G_{K^{perf}}$ is a scheme. Let $K \subset K^{perf} \subset \overline{K}$ be the algebraic closure of K. We may choose an embedding $\overline{k} \to \overline{K}$ over k, so that $G_{\overline{K}}$ is the base change of the scheme $G_{\overline{k}}$ by $\overline{k} \to \overline{K}$. By Varieties, Lemma 0479 we see that $G_{\overline{K}}$ is a Jacobson scheme all of whose closed points have residue field $\overline{K}$.

Since $G_{\overline{K}} \to G_{K^{perf}}$ is surjective, it suffices to show that the image $g \in |G_{K^{perf}}|$ of an arbitrary closed point of $G_{\overline{K}}$ is in the schematic locus of G_K. In particular, we may represent g by a morphism $g : \mathrm{Spec}(L) \to G_{K^{perf}}$ where L/K^{perf} is separable algebraic (for example we can take $L = \overline{K}$). Thus the scheme

$$\begin{aligned} T &= \mathrm{Spec}(L) \times_{G_{K^{perf}}} G_{\overline{K}} \\ &= \mathrm{Spec}(L) \times_{\mathrm{Spec}(K^{perf})} \mathrm{Spec}(\overline{K}) \\ &= \mathrm{Spec}(L \otimes_{K^{perf}} \overline{K}) \end{aligned}$$

is the spectrum of a $\overline{K}$-algebra which is a filtered colimit of algebras which are finite products of copies of $\overline{K}$. Thus by Groupoids, Lemma 0B7U we can find an affine open $W \subset G_{\overline{K}}$ containing the image of $g_{\overline{K}} : T \to G_{\overline{K}}$.

Choose a quasi-compact open $V \subset G_{K^{perf}}$ containing the image of W. By Spaces over Fields, Lemma 0B84 we see that $V_{K'}$ is a scheme for some finite extension K'/K^{perf}. After enlarging K' we may assume that there exists an affine open $U' \subset V_{K'} \subset G_{K'}$ whose base change to $\overline{K}$ recovers W (use that $V_{\overline{K}}$ is the limit of the schemes $V_{K''}$ for $K' \subset K'' \subset \overline{K}$ finite and use Limits, Lemmas 01Z4 and 01Z6). We may assume that K'/K^{perf} is a Galois extension (take the normal closure Fields, Lemma 09DT and use that K^{perf} is perfect). Set $H = \mathrm{Gal}(K'/K^{perf})$. By construction the H-invariant closed subscheme $\mathrm{Spec}(L) \times_{G_{K^{perf}}} G_{K'}$ is contained in U'. By Spaces over Fields, Lemmas 0B86 and 0B87 we conclude. □